Graphic Era HILL UNIVERSITY

End Term (Even) Semester Regular/Back Examination May 2026

Roll no. 2418159

Name of the Program and semester: BTech CSE IV
Name of the Course: Fundamentals of Statistics and AI
Course Code: TCS 421
Time: 3 hours

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Explain the components of a problem in Artificial Intelligence and describe the concept of State Space Representation with a suitable example. (CO3)

b. Define an Intelligent Agent. Explain the anatomy and structure of an intelligent agent and discuss the different types of intelligent agents with examples.(CO1)

c. Illustrate the foundations of AI and discuss the current trends and major applications of Artificial Intelligence in different fields with suitable examples. (CO4)

Q2. (2X10=20 Marks)

a. Define the concept of a Problem Solving Agent in Artificial Intelligence. Describe the steps involved in formulating a problem, including initial state, state space, actions, transition model, goal test, and path cost. (CO1)

b. Compare Uninformed Search Algorithms, Informed Search Algorithms, and Local Search Algorithms in Artificial Intelligence. Discuss their characteristics, performance measures, and applications in solving complex optimization problems. (CO4)

c. Discuss the Uninformed (Blind) Search Methods used in Artificial Intelligence. Explain the working principles, advantages, and limitations of the following algorithms with examples (CO2):

- Breadth First Search (BFS)
- Depth First Search (DFS)
- Depth First Iterative Deepening (DFID)

Q3. (2X10=20 Marks)

a. Describe how does Data Science differ from Business Intelligence? Illustrate the comparison with suitable examples. (CO2)

b. Illustrate data formatting, exploratory data analysis (EDA), filtering, and hierarchical indexing using Pandas. How do these techniques help in preparing data for analysis and visualization? (CO4)

c. Discuss the use of Pandas for data manipulation and the role of visualization tools such as Seaborn and map plotting techniques for presenting analytical results effectively. (CO2)

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Q4.

a. Describe the role of probability distributions in modelling uncertainty and how models are evaluated using statistical methods. (CO2)

b. Explain the steps involved in selecting a model, estimating parameters, and evaluating the model using statistical methods. Also illustrate how R functions and packages can be used for data manipulation, processing, and model fitting. (CO3)

c. Discuss measures such as mean, median, variance, and standard deviation. Demonstrate how these statistical measures can be calculated and interpreted using R programming along with examples. (CO2)

(2X10=20 Marks)

Q5.

a. Compare Linear Regression, Logistic Regression, and Exponential Regression in terms of their mathematical models, assumptions, and applications. Provide examples of situations where each regression model is appropriate. (CO4)

b. Explain the concept of Analysis of Variance (ANOVA) and its importance in statistical analysis. Describe how ANOVA, regression analysis, probability distributions, and correlation techniques collectively help in interpreting data and making statistical decisions in practical applications. (CO3)

c. Illustrate how correlation and covariance are used to measure relationships between variables, and explain their role in linear regression analysis with suitable examples. (CO4)